

HARSTAD, Norway

WMO: 011800

Lat: 68.802N

Lon: 16.537E

Elev: 45

StdP: 100.79

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.2	-9.0	-16.5	0.9	-8.7	-15.2	1.0	-7.9	9.4	-2.0	7.8	-1.8	1.7	210	0.647

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.3	21.7	15.1	19.9	14.4	18.1	13.6	16.2	20.5	15.0	18.9	14.0	17.3	1.7	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.1	10.1	18.3	13.1	9.4	17.2	12.1	8.8	16.0	45.6	20.6	42.1	19.1	39.4	17.3	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.0	5.8	4.9	-12.5	25.1	2.0	3.2	-13.9	27.4	-15.1	29.2	-16.2	31.0	-17.6	33.3		
(5)			-13.4	17.3	1.8	1.8	-14.7	18.6	-15.7	19.7	-16.7	20.7	-18.0	21.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-2.3	-2.4	-0.9	2.6	7.0	9.9	13.4	12.6	9.3	4.8	1.3	0.0
	DBStd	6.35	3.61	3.78	3.39	2.98	3.32	3.00	3.06	2.67	2.83	3.31	3.14	3.19
	HDD10.0	2219	382	346	337	223	104	37	3	6	46	164	261	311
	HDD18.3	4996	641	579	595	473	351	252	157	177	271	419	511	569
	CDD10.0	270	0	0	0	1	12	34	108	87	25	3	0	0
	CDD18.3	5	0	0	0	0	0	1	4	1	0	0	0	0
	CDH23.3	34	0	0	0	0	0	2	28	4	0	0	0	0
	CDH26.7	6	0	0	0	0	0	0	5	1	0	0	0	0
Wind	WSAvg	2.3	2.5	2.4	2.5	2.3	2.4	2.4	2.0	1.9	1.9	2.2	2.2	2.4
Precipitation	PrecAvg	993	100	98	96	60	53	45	62	68	95	117	88	119
	PrecMax	1178	204	198	203	126	95	76	129	156	196	184	186	269
	PrecMin	771	19	17	27	8	15	7	15	34	23	30	0	8
	PrecStd	120	51	44	49	31	24	17	34	27	46	42	54	68
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.5	6.7	8.6	13.2	20.5	22.4	26.7	22.6	17.9	14.2	10.4	8.0
		MCWB	3.9	4.5	3.8	7.1	11.8	14.2	18.6	15.8	12.3	9.8	7.8	5.6
	2%	DB	4.9	5.3	6.3	10.7	17.2	19.3	22.7	20.3	16.2	12.3	8.2	6.7
		MCWB	3.0	3.1	3.3	6.2	10.8	12.9	16.0	15.2	12.1	8.7	6.0	4.6
	5%	DB	3.7	4.2	4.9	8.3	14.2	16.7	20.6	18.5	14.7	10.9	6.7	5.3
		MCWB	1.9	2.4	2.4	4.9	8.9	11.7	15.3	14.3	11.3	8.4	4.5	3.2
	10%	DB	2.5	3.0	3.9	6.9	12.1	14.9	18.8	17.0	13.5	9.6	5.5	4.2
		MCWB	0.8	1.3	1.8	3.9	7.7	10.8	14.3	13.3	10.7	7.3	3.7	2.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	4.8	4.6	7.7	12.7	15.1	18.8	17.1	13.5	10.2	8.2	6.0
		MCDB	5.7	6.5	7.1	12.2	18.9	21.0	25.4	20.3	16.5	13.1	9.9	7.5
	2%	WB	3.2	3.4	3.7	6.4	11.0	13.6	17.0	15.7	12.6	9.4	6.3	4.9
		MCDB	4.7	5.0	5.8	10.1	16.6	18.1	21.9	19.5	15.2	11.8	7.9	6.4
	5%	WB	2.1	2.5	2.8	5.2	9.4	12.2	15.7	14.7	11.7	8.5	4.8	3.5
		MCDB	3.5	4.0	4.7	8.0	13.9	16.1	19.7	17.8	14.1	10.6	6.2	5.0
	10%	WB	1.0	1.5	1.9	4.1	8.1	11.1	14.5	13.7	10.9	7.4	3.8	2.4
		MCDB	2.3	2.8	3.7	6.6	11.7	14.5	18.2	16.5	13.3	9.4	5.3	3.8
Mean Daily Temperature Range	5% DB	MDBR	3.9	4.2	4.9	5.6	6.1	5.7	6.3	5.9	5.4	4.1	3.7	3.9
		MCDBR	4.5	4.8	4.9	7.0	9.5	8.9	9.8	8.3	6.7	4.9	4.3	4.8
		MCWBR	3.8	4.0	3.5	4.4	5.2	4.9	4.9	4.4	4.0	3.5	3.4	4.1
	5% WB	MCDBR	4.2	4.7	4.8	6.6	9.2	8.3	9.2	7.6	6.1	4.8	4.1	4.8
		MCWBR	3.8	4.1	3.6	4.5	5.2	4.9	4.9	4.4	3.9	3.8	3.5	4.3
Clear-Sky Solar Irradiance	taub	0.372	0.268	0.308	0.329	0.339	0.340	0.344	0.334	0.317	0.278	0.351	N/A	
	taud	2.653	2.116	2.209	2.326	2.444	2.498	2.515	2.531	2.542	2.261	2.620	N/A	
	Ebn at Noon	93	491	706	814	852	862	844	812	728	483	78	0	
	Edh at Noon	11	44	76	91	91	89	85	75	57	37	9	0	
All-Sky Solar Radiation	RadAvg	0.02	0.30	1.36	3.15	4.72	4.90	4.41	3.15	1.68	0.55	0.06	0.00	
	RadStd	0.00	0.06	0.22	0.28	0.38	0.66	0.46	0.29	0.18	0.10	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)		+0.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page